BLUE LOTUS RESEARCH INSTITUTE

PhD-Level Investment Due Diligence — Materials — Aluminium

Alcoa Corporation

[AA] — Materials — Aluminium Due Diligence

Aluminium Turnaround — $13.1B Market Cap · 10.3x P/E · EV Lightweighting Tailwind · Alumina Vertical Integration · Energy Cost Reset

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SECTION 1 — EXECUTIVE SUMMARY

Verdict: Buy | Conviction: 7.9/10

Alcoa Corporation (AA) — Materials — Aluminium PhD Due Diligence. All prices and financials from BL3 MySQL (bluelotus3.historical_prices, ticker_fundamentals). Zero Claude training-data figures. Verdict: Buy. Conviction: 7.9/10.

Metric	Value
Price (BL3, 2026-08-07)	$50.17
Market Cap	$13.1B
P/E (TTM)	10.31x
P/B	1.78x
EPS (TTM)	$4.37
Net Income (TTM)	$1.15B
Dividend Yield	0.81%
52-Week High	$84.22
52-Week Low	$28.69

SECTION 2 — COMPANY OVERVIEW, BULL & BEAR CASE

Bull Case:

- EV lightweighting structural tailwind: aluminium is 60-70% lighter than steel at equivalent strength — every electric vehicle uses 2-3x more aluminium than an internal combustion vehicle (battery enclosures, structural components, heat management); the EV manufacturing ramp is a decade-long demand driver for aluminium
- Vertically integrated from bauxite to aluminium: Alcoa controls bauxite mines (Australia, Brazil), alumina refineries, and aluminium smelters — vertical integration provides cost stability and margin protection through the commodity cycle; peers who lack alumina assets face supply chain risk
- 10.3x P/E at cyclical trough: Alcoa has historically traded at 6-15x through the cycle — current pricing is at the lower half of the historical range despite EV demand tailwinds; the cycle is turning with China stimulus and automotive demand recovery
- Warrick rolling mill divestiture: Alcoa divested non-core rolling operations to focus on primary aluminium — the simplified portfolio improves operating leverage and capital allocation clarity
- Energy cost structure improvement: Alcoa closed high-cost European smelters (San Ciprián, Lista) that were loss-generating in high energy cost environments — remaining global smelter portfolio is weighted toward lower-energy-cost jurisdictions (Iceland, Australia, Canada)

Bear Case:

- Energy cost sensitivity: aluminium smelting is one of the most electricity-intensive industrial processes — energy represents 30-40% of smelting cash cost; electricity price spikes in Europe and North America can render smelters uneconomic rapidly
- China aluminium overcapacity: China accounts for 60%+ of global aluminium production and has historically exported surplus at below-cost prices — Chinese overcapacity export waves compress global aluminium prices and margins for non-Chinese producers
- San Ciprián restart risk: Alcoa suspended the Spanish San Ciprián smelter; restart obligations create a fixed cost burden without corresponding revenue if energy prices remain elevated in Spain

- Commodity price cyclicality: aluminium spot price ranged from $2,100 to $3,600/tonne in 2022-2024 — Alcoa's earnings are highly leveraged to aluminium price; a $100/tonne price move equates to ~$150-200M in annual earnings impact
- Newcore Acquisition integration: Alcoa acquired Alumina Limited — creating a 100% owned Alumina interest; while strategically sound, integration adds complexity and short-term transaction cost

SECTION 3 — FINANCIAL PERFORMANCE (BL3 MySQL)

Financial Metric (BL3 MySQL)	Value
P/E (TTM)	10.31x
Forward P/E	11.35x
P/B	1.78x
Net Income (TTM)	$1.150B
EPS (TTM)	$4.37
Market Cap	$13.1B
Shares Outstanding	263M
Dividend Yield	0.81%
52-Week High	$84.22
52-Week Low	$28.69
Price (BL3, 2026-08-07)	$50.17

SECTION 4 — PORTER'S FIVE FORCES (MATERIALS — ALUMINIUM)

Force	Intensity	Analysis
Rivalry	MEDIUM-HIGH	Iron ore (VALE vs Rio Tinto/BHP), aluminium (AA vs Alcoa/Rusal/Chalco), and steel (CLF/NUE vs domestic mills) are intensely competitive on price; commodity businesses compete primarily on cost of production rather than product differentiation. Fertiliser (NTR/MOS) is a global oligopoly driven by nutrient scarcity and Canpotex export cartel (NTR/Mosaic/Nutrien control 30%+ of global potash). ADM competes with Cargill, Bunge, and Louis Dreyfus in a global grain processing oligopoly. CL (consumer staples) competes in a rational duopoly/oligopoly (Procter & Gamble, Unilever). OXY competes against majors (Exxon, Chevron) in US Permian — cost curves determine competitive position. SentinelOne (S) competes intensely with CrowdStrike, Microsoft Defender, Palo Alto in EDR/XDR.
New Entrants	LOW	Mining requires $5-20B in capital, multi-decade permitting, and geological assets that cannot be created; no new US integrated steel mill has been built in 30+ years. Fertiliser mining (potash, phosphate) requires access to geological deposits that are geographically concentrated (Canada, Morocco, Russia, China). ADM's grain elevator and processing network took a century to build. Colgate's brand (120+ year trust in 200 countries) cannot be replicated. Oil & gas requires access to acreage, drilling expertise, and midstream infrastructure. Cybersecurity (SentinelOne) has lower capital barriers but requires AI model training data at scale — 8M+ endpoints provides an increasingly hard-to-replicate training advantage.
Supplier Power	MEDIUM	Mining companies control their own deposits (VALE owns Carajás, world's largest iron ore mine) — own supplier position.

		Steel mills (CLF, NUE) are buyers of scrap metal and iron ore; scrap prices are market-set with moderate supplier leverage. NTR controls Potash Corp legacy mines but nitrogen requires natural gas (Henry Hub price-driven). ADM buys grains from farmers at commodity prices — no single farmer has pricing power. OXY operates in the Permian where service company costs (Halliburton, SLB) carry moderate leverage during drilling activity peaks. SentinelOne buys cloud compute (AWS/Azure) — standard market pricing.
Buyer Power	MEDIUM	Steel buyers (automakers, construction) are sophisticated procurement operations that negotiate volumes annually — some switching between domestic mills (CLF, NUE, STLD) based on price. Agricultural buyers (grain traders, food processors) purchase ADM's output at commodity prices. Consumer goods buyers (supermarkets, mass market retailers) negotiate hard with CL — private label is the credible threat. Oil buyers (refiners) buy at spot — no single buyer commands pricing power. Cybersecurity buyers are large enterprises — budget cycles are multi-year but competitive bids occur at contract renewal; Microsoft's bundled Defender creates perpetual competitive pressure on SentinelOne.
Substitutes	MEDIUM	Aluminium substitutes steel in automotive (AA benefits); carbon fibre substitutes both at lower scale. Scrap steel recycling substitutes virgin iron ore in electric arc furnaces (NUE's model); VALE is exposed to DRI/HBI adoption. Potash has no substitute for crop nutrition — agricultural yield loss from potassium deficiency is immediate and severe. Alternative proteins (insects, lab meat) are very long-term ADM substitutes. Consumer staples substitutes are private label — Colgate has resisted private label

		in oral care more than most CPG peers. Natural gas substitutes coal for power (positive for OXY's production mix). AI-powered security substitutes or supplements third-party EDR — Microsoft Copilot for Security is the largest substitute risk for SentinelOne.

SECTION 5 — KEY RISKS

- Commodity price cyclicality: metals (VALE, AA, CLF, NUE), fertilisers (NTR, MOS), and oil (OXY) all experience multi-year price cycles driven by global supply-demand imbalances; earnings can collapse 60-80% from peak to trough in a single cycle
- China demand dependency: iron ore (VALE), aluminium (AA), potash (NTR), and phosphate (MOS) are all heavily dependent on Chinese agricultural and industrial demand — any Chinese economic slowdown creates immediate volume and price pressure
- Geopolitical supply disruption risk: Russia/Belarus potash sanctions, Middle East oil supply disruptions, and South American mining political risk all create potential supply-demand shocks in opposite directions across the portfolio
- Carbon transition and decarbonisation pressure: steel (CLF, NUE), fertiliser production, oil (OXY), and grain processing (ADM) are carbon-intensive industries facing increasing regulatory pressure, carbon taxes, and investor ESG mandates
- Cybersecurity competition (SentinelOne): Microsoft's bundled Defender for Endpoint, CrowdStrike's market recovery, and Palo Alto's XSIAM platform all represent credible competitive threats; the AI security market is evolving rapidly and the winner is not yet determined
- ADM accounting investigation: ongoing SEC investigation into ADM's nutrition segment intersegment transactions creates headline risk and potential earnings restatement; management credibility has been partially impaired

SECTION 6 — INVESTMENT THESIS & VERDICT

VERDICT: BUY | CONVICTION: 7.9/10

Alcoa Corporation (AA) — Buy with 7.9/10 conviction. BL3 data: Price $50.17, MCap $13.1B. Materials, energy, and cybersecurity represent essential industrial infrastructure, food security, and digital security — each with distinct moats and cyclical characteristics.

Data: BL3 MySQL bluelotus3.ticker_fundamentals + historical_prices | CivilisationOS RAG (WSJ/FT/NIKKEI). Work Order: WO-DD-MEGA-MATH-20260814. Report Date: 14 August 2026. Zero training-data figures. AMP compliant.